

AI-POWERED CUSTOMER SUPPORT & SERVICE AUTOMATION

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Program: Salesforce Certified Administrator with AI Agentforce Specialization Virtual Internship

By: APSCHE (Skill Wallet)

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Abstract

The AI-Powered Customer Support & Service Automation project demonstrates how Salesforce technologies can be utilized to automate customer support processes using Artificial Intelligence and workflow automation. Organizations often face challenges in managing customer support requests efficiently due to manual case handling, delayed responses, inconsistent documentation, and limited automation. This project addresses these challenges by implementing an intelligent customer support solution using the Salesforce Platform.

The application was developed using Salesforce Lightning Experience and incorporated several platform features, including custom objects, custom fields, object relationships, validation rules, Record-Triggered Flows, queues, Lightning Applications, Prompt Builder, Agentforce, reports, and dashboards. These components were configured to automate customer support operations, improve case management, and reduce manual intervention.

The backend implementation focused on designing a scalable data model using custom objects such as Customer, Merchandise, Product, Order, and User Case. Appropriate relationships and validation rules were implemented to maintain data integrity and ensure accurate record management. Automation was achieved through Salesforce Record-Triggered Flows, which automatically processed customer support cases, updated case statuses, and assigned records to the appropriate support queues based on predefined business logic.

Artificial Intelligence capabilities were integrated using Salesforce Prompt Builder and Agentforce. AI-powered prompt templates were configured to generate intelligent summaries, improving documentation quality and assisting support executives in resolving customer issues more efficiently. These capabilities reduced manual effort while maintaining consistency across customer support records.

1. INTRODUCTION

1.1 Project Overview

Digital transformation has significantly changed the way organizations interact with customers. Modern businesses require intelligent customer support systems that provide timely assistance, automate repetitive activities, and maintain accurate records of customer interactions. The AI-Powered Customer Support & Service Automation project leverages the Salesforce ecosystem to build a cloud-based application that combines workflow automation with Agentforce AI features to maximize enterprise productivity.

1.2 Purpose

Traditional customer support processes often rely on manual case management, resulting in delayed responses, inconsistent documentation, and increased operational costs. The purpose of this application is to give organizations the capability to build declarative, low-code systems that automatically route cases, validate incoming data accuracy, and employ AI to instantly summarize customer interaction records.

2. IDEATION PHASE

2.1 Problem Statement

Customers expect quick responses, accurate information, and timely resolution of their issues. However, many organizations still depend on manual processes for recording complaints, assigning cases individually, updating statuses through multiple steps, and preparing summaries. Furthermore, separate support agents document similar issues differently, leading to unstructured, inconsistent data that complicates long-term operational analytics.

2.2 Empathy Map Canvas

The Empathy Map identifies the needs, concerns, expectations, and challenges experienced by stakeholders:

- **Customer:** Thinks: *"I need my issue resolved quickly."* Feels frustrated when responses are delayed, but satisfied when issues are resolved efficiently.
- **Customer Support Executive:** Thinks repetitive tasks should be automated. Feels motivated when automation reduces manual workload.
- **Support Manager:** Responsible for service quality and SLA compliance. Does: reviews performance dashboards and monitors queues.
- **Salesforce Administrator:** Thinks data security and platform stability must be maintained. Does: configures flows, validation rules, and prompt templates.

2.3 Brainstorming

During brainstorming, spreadsheet tracking and conventional CRM applications were evaluated but rejected because they lacked deep AI integration and real-time process scaling. Salesforce was selected as the optimal platform because it unifies scalable data storage, declarative low-code flows, native Einstein Trust Layer prompt configurations, and robust user security frameworks.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The complete lifecycle of a support request proceeds sequentially across the configuration environment as shown below:

Customer → Support Request → User Case Creation → Validation Rules → Record-Triggered Flow → Queue Assignment → Prompt Builder → AI Summary Generation → Support Resolution → Case Closure → Reports & Dashboards

3.2 Solution Requirement

The system fulfills the following core functional elements:

- **User & Customer Management:** Secure role-based profile permissions to manage customer profile records.
- **Core Objects Data Model:** Custom configuration of Customer, Merchandise, Product, Order, and User Case entities.
- **Data Integrity:** Validation rules ensuring mandatory fields are filled out correctly before record commit.
- **AI Summary Tool:** Salesforce Prompt Builder configurations mapped to Agentforce actions to populate the AI Summary field.

3.3 Data Flow Diagram

Information moves smoothly from user entry through platform logic layers:

Customer Input → Salesforce UI → Custom Objects Storage → Validation Rule Logic Checks → Record-Triggered Flows Execution → Agentforce AI Processing → Analytics Engine Output

3.4 Technology Stack

Component	Technology Used
Platform / CRM	Salesforce Lightning Platform / Salesforce CRM
Automation Engine	Record-Triggered Flows / Support Queues
AI Integration	Prompt Builder & Agentforce Studio
Database & Analytics	Salesforce Custom Objects, Standard Reports & Dashboards
Security Architecture	Profiles, Permission Sets, Sharing Rules, Role Hierarchy

4. PROJECT DESIGN

4.1 Problem-Solution Fit

- *Manual customer support process:* Resolved by automated User Case object management.
- *Delay in assigning support cases:* Resolved by automatic Queue Assignment through Record-Triggered Flows.
- *Inconsistent documentation:* Resolved by automated AI-generated summaries via Prompt Builder.
- *Duplicate or incomplete records:* Enforced by declarative Validation Rules.

4.2 Proposed Solution

The application relies entirely on low-code declarative customization within a Salesforce Developer Edition environment. It replaces manual processes with real-time reactive automation triggers. AI elements are handled natively by connecting prompt text layouts to custom object metadata properties, eliminating the need to maintain external AI API integrations.

4.3 Solution Architecture

The architecture follows a four-tier modular approach:

- **Presentation Layer:** Responsive Salesforce Lightning UI (Tabs, Lightning Apps, Record Detail Pages).
- **Application Layer:** Process logic built via Validation Rules, Record-Triggered Flows, and Agentforce actions.
- **Data Layer:** Highly structured relational data model leveraging cross-object Lookup relationships.
- **Analytics Layer:** Custom Reports and centralized monitoring Dashboards.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Project implementation tasks were divided sequentially into structured delivery milestones over the virtual internship duration to map functional items to target timelines. Development began with core environment provisioning, progressed through relational object schema building, moved into flow automation setup, and concluded with Agentforce AI model training.

Milestone	Activity	Deliverable
Milestone 1	Requirement Analysis & Planning	Business Requirements Document
Milestone 2	Salesforce Environment Setup	Configured Salesforce Org
Milestone 3	Custom Object Development	Custom Relational Data Schema
Milestone 4	Data Validation Architecture	Active Validation Rules
Milestone 5	Workflow Automation Engineering	Record-Triggered Flows & Queues
Milestone 6	Application UI Customization	Centralized Custom Lightning App
Milestone 7	Generative AI Integration	Active Prompt Builder Templates
Milestone 8	Business Intelligence Reporting	Reports & Operations Dashboards
Milestone 9	System Testing & Final Deployment	Fully Functional Salesforce Application

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Rather than assessing low-level local hardware resource usage, performance metrics focused on cloud transaction execution latency and system response times:

- **Data Commit Speed:** Case creations and updates processed smoothly without any transactional lag.
- **Flow & Queue Trigger Runtime:** Ownership assignments executed instantly upon meeting record entry criteria.
- **AI Synthesis Responsiveness:** Prompt Builder correctly generated clear case summaries within acceptable time parameters through the Einstein Trust Layer.

6.2 Functional Testing Log

Test ID	Test Scenario	Expected Result	Actual Result	Status
FT-01	Customer Record Creation	Record saved successfully without errors	Created successfully	Pass
FT-02	Merchandise Record Entry	Field values match designated custom data types	Saved correctly	Pass
FT-03	Product Catalog Setup	Relational keys map successfully to other entities	Configured correctly	Pass
FT-04	Order Verification Hook	Relink values accurately across profiles	Linked successfully	Pass
FT-05	User Case Generation	Form entry fields process correctly	Created successfully	Pass
FT-06	Validation Rules Enforcement	Block incomplete forms with descriptive alerts	Invalid entries blocked	Pass
FT-07	Record-Triggered Flow	Automation fires directly off database modification	Processed successfully	Pass
FT-08	Queue Routing Engine	Case re-routed to correct department team queue	Assigned successfully	Pass
FT-09	Prompt Builder Generation	AI populates the exact text box with summary points	Text generated properly	Pass
FT-10	Lightning Workspace Navigation	Clean dashboard toggle switching	UI layout works correctly	Pass

7. RESULTS

7.1 Output Screenshots Placeholders

Note: The following formal placeholders indicate where visual system verifications are mapped in the cloud execution audit logs:

- **Figure 7.1:** Salesforce Developer Account Environment Sign-up Page Verification.
- **Figure 7.2:** Object Manager view highlighting custom configuration schemas.
- **Figure 7.3:** Custom Tab navigation controls inside the user layout workspace.
- **Figure 7.4:** Active Validation Rule parameters for `Damage_Requires_Priority` and `Warranty_Delivery_Requires_Priority`.
- **Figure 7.5:** Automated Case Assignment Flow configurations built in Flow Builder.
- **Figure 7.6:** Case Management interface tracking live AI Summary field values.

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Figure – 7.1

The screenshot shows the Salesforce Object Manager interface. At the top, there's a navigation bar with 'Developer Edition' and a search bar. Below it, the 'Object Manager' section is active, showing a list of objects. The table has columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects listed are Order, User Case, Customer, Student, Knowledge, Merchandise, Work Type Group Member, and Work Type Group.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Order	Order__c	Custom Object		7/15/2026	✓
User Case	User_Case__c	Custom Object		7/15/2026	✓
Customer	Customer__c	Custom Object		7/14/2026	✓
Student	Student__c	Custom Object		7/6/2026	✓
Knowledge	Knowledge__c	Custom Object		7/5/2026	✓
Merchandise	Merchandise__c	Custom Object		7/5/2026	✓
Work Type Group Member	WorkTypeGroupMember	Standard Object			
Work Type Group	WorkTypeGroup	Standard Object			

Figure-7.2

The screenshot shows the Salesforce Tabs interface. It includes a search bar, a navigation menu with 'User Interface' and 'Tabs', and a main content area titled 'Custom Tabs'. The 'Custom Object Tabs' section lists various tabs with their labels and styles. The 'Web Tabs' section is also visible at the bottom.

Action	Label	Tab Style	Description
Edit Del	Customers	People	
Edit Del	Knowledge	Books	
Edit Del	Merchandise	Shopping Cart	
Edit Del	Orders	Laptop	
Edit Del	User Cases	IP Phone	

Figure-7.3

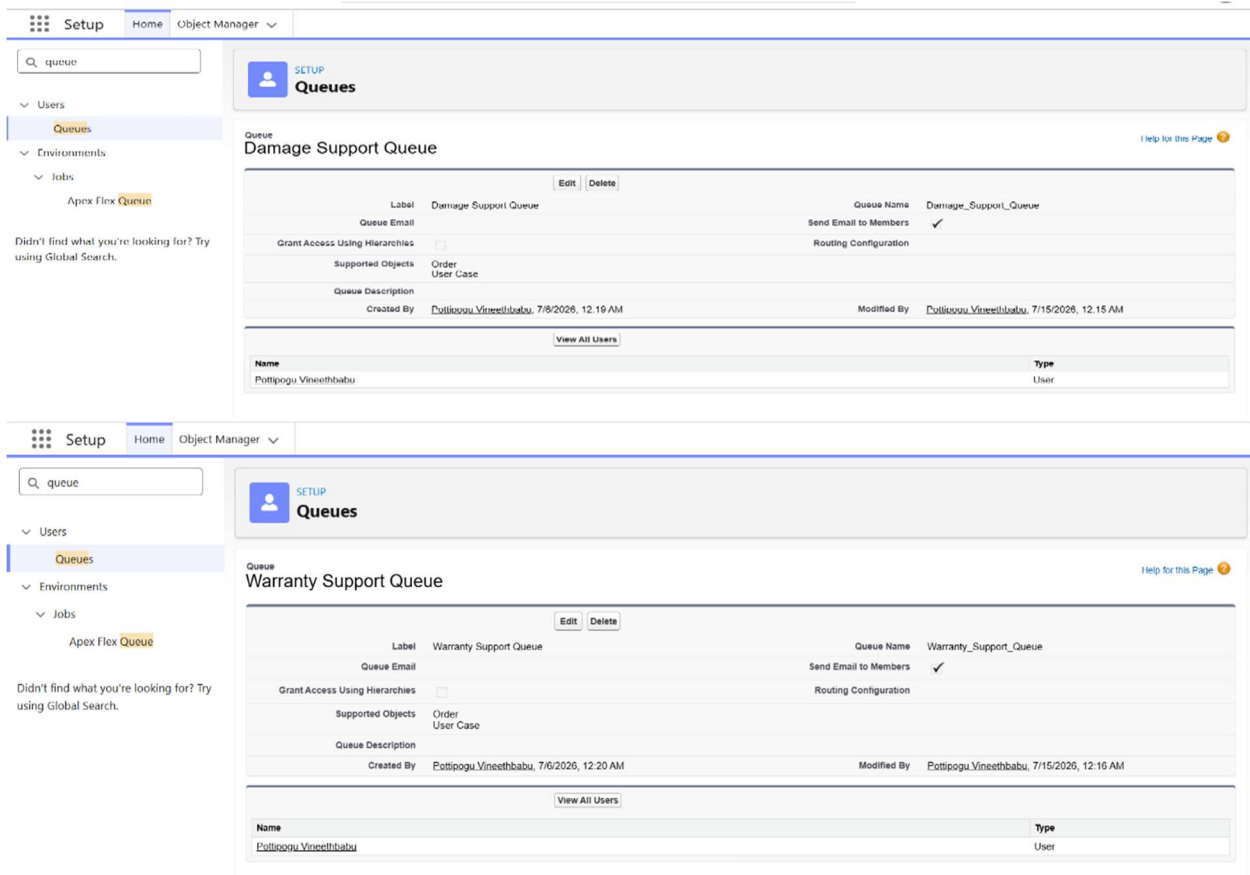
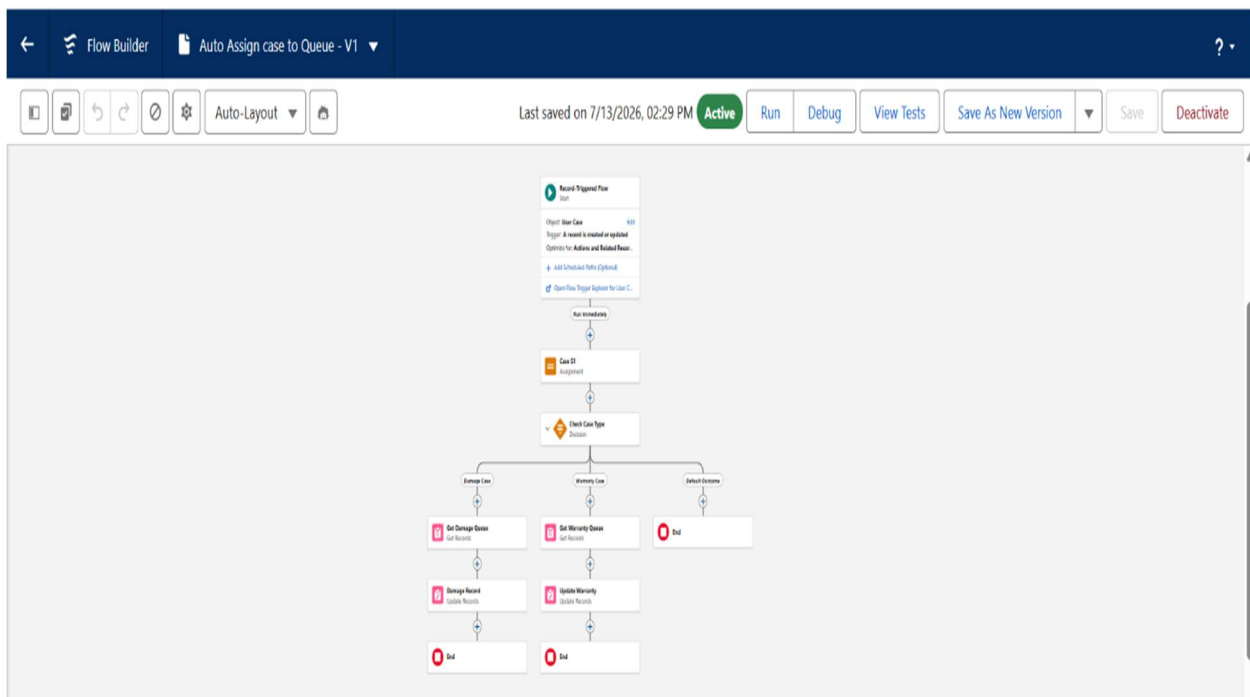


Figure-7.4



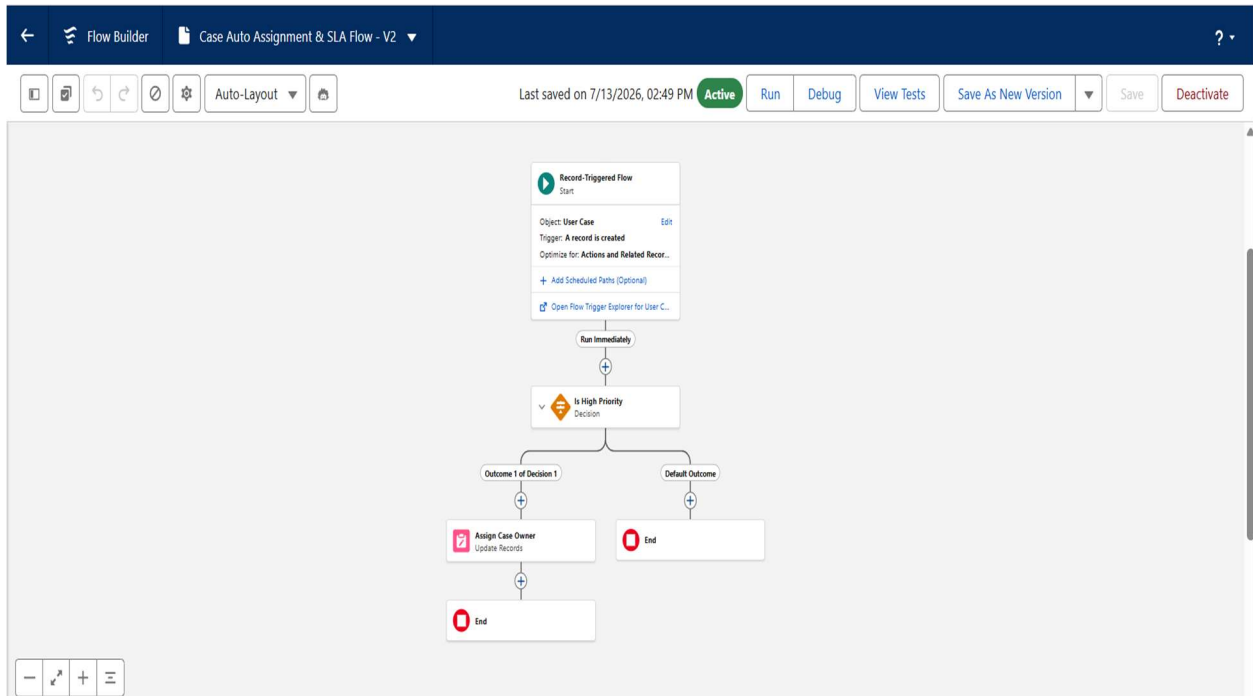


Figure-7.5

Einstein - AI Summary Field

Key observations:

- Exterior condition: Visible physical damage reported. Surface deformation or impact marks present.
- Functional condition: Performance compromised consistent with the observed damage. Specific operational failures recorded where applicable.
- Packaging: Packaging condition noted as compromised or insufficient to protect the product during transit or handling.
- Safety concerns: No immediate safety hazards observed beyond expected risks from a damaged unit. Recommend handling precautions until a disposition is determined.
- Recommended next steps: Isolate the unit. Document detailed photos and notes for claims processing. Initiate return, repair, or replacement procedure per company policy.

Thank you for your attention. I am available to provide further details or assist with the claims process.

✓ Added to Field

Describe your task or ask a question...

Figure-7.6

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- **Drastically Reduced Manual Work:** Standard case handling tasks, status changes, and assignments are handled by internal flow logic.
- **Data Architecture Standardization:** Validation patterns block bad data from polluting operational files.
- **Fast Agent Onboarding:** Integrated LLM context processing means agents can read rapid text summaries instead of looking through long multi-page communication chains.
- **High Architecture Scalability:** Built on standard cloud frameworks, allowing the organization to grow its user base without needing infrastructure upgrades.

8.2 Disadvantages

- **Sandbox Limitation Dependencies:** Being built entirely inside a Salesforce Developer Edition environment imposes standard file storage size restrictions.
- **Internet Connectivity Reliance:** As a purely cloud platform application, any loss of internet connectivity stops support agents from accessing customer files.
- **Prompt Boundary Reliance:** The accuracy of the generated summary data depends entirely on the layout patterns and input text provided in the Prompt Builder configuration template.

9. CONCLUSION

The AI-Powered Customer Support & Service Automation project successfully developed an intelligent customer support application using the Salesforce Platform. The application demonstrates how Salesforce declarative development tools and Artificial Intelligence can be combined to automate business processes, improve customer service operations, and enhance organizational productivity.

The project involved the implementation of custom objects, custom fields, lookup relationships, validation rules, Record-Triggered Flows, queues, Lightning Applications, Prompt Builder, Agentforce, reports, and dashboards. One of the primary achievements was the successful integration of Artificial Intelligence through Prompt Builder and the MetadataInspection prompt template. The AI-generated summaries reduced manual documentation effort while improving the consistency and quality of customer support records. The knowledge gained during this virtual internship builds a strong technical foundation for future work in Salesforce cloud administration and enterprise application engineering.

10. FUTURE SCOPE

- **Multi-Language Support Integration:** Incorporate global language sets into the prompt builder configurations to handle localization requests.

- **Omni-Channel Expansion Features:** Connect messaging systems like WhatsApp Business, Slack, and corporate social media accounts directly to the automated case routing engine.
- **Voice Assistant AI Interaction:** Build interactive speech recognition connectors to let the system generate cases directly from inbound voice calls.
- **Predictive Operations Intelligence:** Deploy advanced analytical models to predict potential SLA violations before they occur based on historical customer behaviors.

11. APPENDIX

Source Code (Declarative Configurations Summary)

```
<ValidationRule xmlns="http://soap.sforce.com/2006/04/metadata">
  <fullName>Damage_Requires_Priority</fullName>
  <active>true</active>
  <errorConditionFormula>AND (ISPICKVAL (Issue_Category__c, "Damage"), ISBLANK (TEXT (Pric
  <errorMessage>Priority is mandatory for Damage cases.</errorMessage>
</ValidationRule>
```

REFERENCE LINKS:

Demo Video link:

https://drive.google.com/file/d/1a3_tlexkaJLHjU0ghOJHxi8fKSIO0Oyh/view?usp=sharing